Enhanced U-Net Architecture for Brain Tumor Localization and Segmentation in T1-Weighted MRI

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***Abstract*—Detection and segmentation of Magnetic Resonance Imaging (MRI) scans is a critical task in medical imaging, where achieving high segmentation precision and reliability remains challenging due to variations in tumor shape, size, intensity, and boundary definition across different MRI modalities. Recently, deep learning techniques have significantly improved the efficiency oflocalization and segmentation of various medical fields, including brain tumoursanalysis. This brief presents a novel two-stage approach for brain tumour segmentation in T1-weighted contrast-enhanced MRI (CE-MRI) scans, leveraging both YOLO (you only look once) and Modified U-Net. In the first stage, YOLO is employed to quickly and accurately localize regions of interest (ROIs) where brain tumours are present. To accelerate this step, YOLOv3 are incorporated, improving the computation speed and efficiency of the model. In the second stage, a modified U-NET model, enhanced with spatial and channel attention modules, is utilized to perform precise segmentation of the tumours within the identified ROIs. The performance of the proposed framework is evaluated using metrics including precision, recall, Jaccard index, and Dice similarity coefficient (DSC). Our approach demonstrates promising results compared to previous methods using the same database.**

***Index Terms*—Brain tumour segmentation, channel attention, spatial attention, U-Net, Yolov3, medical imaging, deep convolutional neural networks.**

I. Introduction

BRAIN tumours are a significant cause of cancer-related mortality globally, affecting individuals across all age groups including pediatric populations [1], [2]. Medical imaging plays an essential role in modern healthcare, where early detection is key to accurate diagnosis and effective treatment planning. Achieving accurate segmentation is important for clinical applications, including brain tumours from MRI scans. To differentiate tumorous regions from healthy brain tissue, brain tumour segmentation involves defining and identifying tumour boundaries in brain images obtained through medical imaging techniques like MRI [3]. In the medical domain, segmentation is usually performed by radiologists. However, manual segmentation can be both time-intensive and prone to human error and challenging for medical experts, since brain tumours, due to their varied shape and size can easily be misinterpreted as other lesions. Consequently, machine learning (ML) and deep learning (DL) techniques have become increasingly popular in recent years, offering more precise, efficient, and automated solutions, particularly in the field of brain tumour segmentation [4]. Hence researchers have relied on deep learning techniques, such as U-Net-based models and object detection frameworks like YOLO (you only look once) [5], to develop automated and reliable systems for accurate brain tumour analysis. Various medical imaging techniques are utilized to capture information about tumours characteristics, including their size, shape, location and types, all of which are essential for accurate diagnosis. Magnetic resonance imaging (MRI), computed tomography (CT), X-ray, and ultrasound provide valuable insights into tumour characteristics, enabling physicians to make informed decisions. MRIs are the most commonly used method among these techniques due to their favourable characteristics.

A literature review shows that a significant number of recent studies in image analysis employ deep convolutional neural networks (CNNs) for tasks such as image classification and segmentation. DL-based approaches have shown exceptional performance in capturing complex spatial patterns and overcoming the limitations of traditional image processing techniques [4], [7], [8]. Advances in model architecture, optimization, and using effective strategies have led to more accurate results in brain tumour segmentation. However, despite the significant advancements achieved by DL-based models, maintaining a balance between improving segmentation performance and managing computational complexity remains a challenging task [9], [10]. The objective is to enhance the precision of brain tumour segmentation while maintaining a balance with computational efficiency. Advances in model design, optimization, and the use of efficient techniques can lead to achieving precise results without significantly increasing computational complexity. The Brain Tumor Segmentation (BRATS) challenge, organized by the University of Pennsylvania and the Centre for Biomedical Image Computing and Analytics (CBICA) since 2012, is a leading competition focused on brain tumour segmentation [11]. With annual updates to its dataset, the performance of algorithms proposed for the challenge has significantly improved over time, achieving a segmentation success rate of around 90% or higher. The BRATS challenge aims to automate brain tumour detection and develop related techniques and algorithms. Another widely used brain tumour dataset is the Figshare MRI brain tumour dataset from 233 patients, including meningiomas, gliomas, and pituitary tumours, which has been employed in this brief [6]. Due to its widespread accessibility and ease of availability, the Figshare MRI dataset has been used in numerous studies on brain tumour classification and segmentation. Various techniques and algorithms have been proposed for this dataset, including those utilizing CNN architectures and advanced CNN models like U-Net [12], [13], [14], [15]. Further information about the dataset is provided in Section II. While CNN-based networks have significantly enhanced performance in medical imaging tasks such as classification and segmentation, they also introduce a

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considerable increase in computational cost, particularly when processing high-resolution MRI scans. In object detection algorithms, objects are typically identified by outlining the ROIs with a bounding box. Traditional object detection models such as R-CNN and Fast R-CNN commonly use a CNN-based feature extractor, followed by fully connected layers for object classification and predict their bounding box. However, these approaches present challenges when the number of detected objects varies, as fully connected layers require a predefined output size during model design. This fixed structure makes it difficult to adapt to images containing an arbitrary number of objects, limiting flexibility in dynamic detection scenarios. Recent detection models, such as YOLO, address this limitation by replacing fully connected layers with anchor based methods for bounding box prediction, enabling more efficient and optimized object detection. This brief presents an automated two-stage deep learning approach for brain tumour segmentation in T1-weighted MRI scans, using YOLOv3 [16] algorithm for accurately localizing ROIs, where brain tumours are present. In the second stage, an enhanced modified U-NET architecture is utilized to perform precise segmentation of the tumours within the identified regions. The key contributions of this brief are summarized as follows:

(1) In the initial phase, the YOLOv3 [16] algorithm is employed to identify and detect the regions of interest (ROIs) within the MRI image. This phase generates the coordinates and creates a bounding box around the tumour area.

(2) The output from YOLOv3 is now used to compute the dot product with the original image. This result is added back to the original image, generating a new input that highlights the tumour boundary within the bounding box. To reduce the computational cost, a patch corresponding to the bounding box, based on the obtained coordinates, is extracted and fed into the U-Net architecture. This enhanced patch is then processed to produce the final segmentation output.

(3) In the subsequent phase, a modified U-Net architecture is introduced to achieve precise segmentation of the identified regions within the enhanced input. This brief explores the benefits of incorporating attention mechanisms, including channel attention and spatial attention modules, into the network to enhance its performance. This approach addresses the limitations of traditional skip connections in standard U-Net models, resulting in improved segmentation performance.

(4) The method is trained and tested using a Figshare MRI dataset, which includes cases of meningiomas, gliomas, and pituitary tumours in common views such as sagittal, coronal, and axial orientations. The proposed methodology demonstrates promising performance when compared to previous techniques that utilized the same database. The remainder of this brief is organized as follows: Section I provides a brief overview of the theoretical background of the research. Section II presents an overview of the proposed framework, methodology, dataset preparation, and network architecture. The experimental results and performance analysis are discussed in Section III, where we also explore the role of pre-processing techniques and data augmentation strategies in improving the generalization and efficiency of our method. The simulation results demonstrate that our model outperforms previous studies conducted on this dataset. Finally, Section IV summarizes the key findings and concludes this brief.

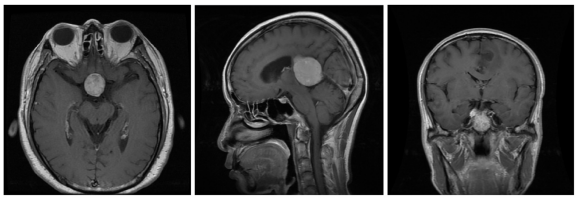

Fig. 1. Different tumour types observed across various planes in dataset [6].

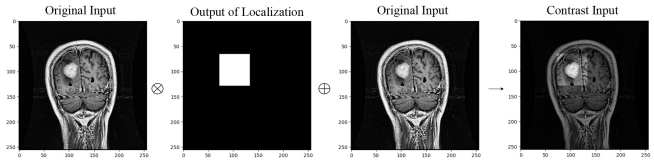

Fig. 2. The output from YOLOv3 is used to compute the dot product with the original image. The resulting combination is added back to the original image, generating a new input that highlights the tumour boundary within a bounding box. This enhanced input is then passed to the U-Net to produce final segmentation.

II. Methodology

This brief proposes an automated approach that integrates the complementary strengths of both U-Net and YOLOv3 models for enhanced brain tumour detection and segmentation using T1-weighted MRI sequences. The proposed architecture consists of two primary stages: utilizing YOLOv3 architecture to detect and classify the tumorous brain tissue from the given input of MRI, followed by the attention-enhanced U-Net architecture which performs precise segmentation. In the first step, the input MRI is pre-processed, resized, and passed through the YOLOv3 algorithm to localize the tumour, generating a bounding box around the tumorous region. This step, referred to as the localization network, provides the coordinates of the bounding box, which encapsulates the tumour within the MRI scan. As illustrated in Figure 2, the output from this step is further processed to create a new input image. This is achieved by performing an element-wise multiplication (dot product) of the original image with the bounding box output and this combined result then will be added to the original image. This process generates a new input image, called the "contrast input," which emphasizes the tumour boundaries, thereby improving the segmentation effectiveness in the subsequent stage. In the next phase, a modified U-Net is implemented as the second network architecture, leveraging the contrast input to generate precise accurate segmentation outputs. The contrast image generated in the previous step is a new input into the modified U-Net architecture, which incorporates both channel and spatial attention blocks, to enhance the model's focus on relevant features throughout the segmentation process. The feature map is divided into two groups along the channel dimension, with spatial and channel attention blocks applied simultaneously to each group. Spatial attention blocks identify ROIs, while channel attention blocks extract meaningful representations from the feature set. This dual attention mechanism addresses the limitations often associated with traditional skip connections in U-Net models, thereby improving segmentation performance. To reduce computational costs, only the image patches corresponding to the bounding box coordinates identified by YOLOv3 are used for training. By focusing on these specific patches, the computational complexity is

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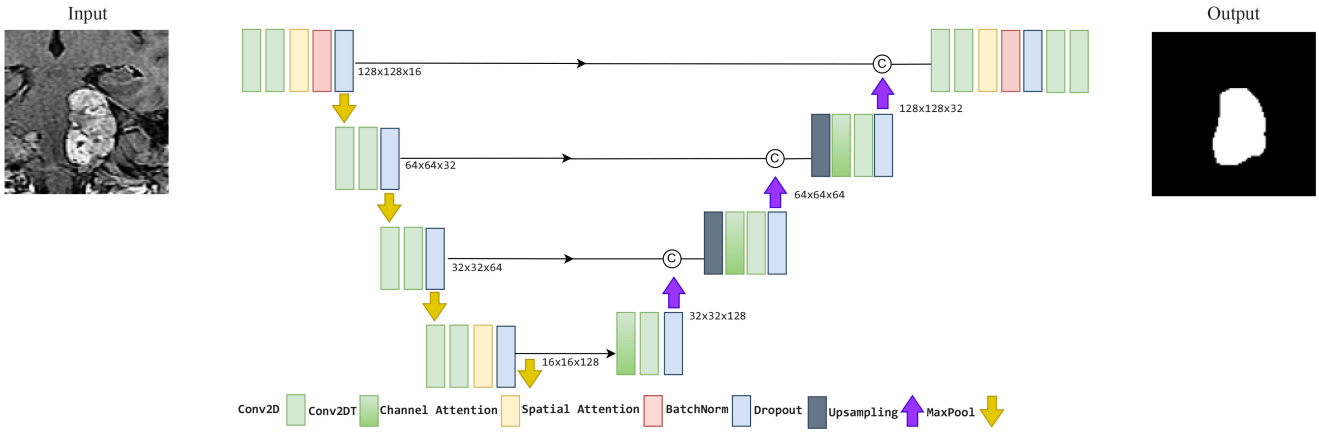

Fig. 3. The visual representation of the proposed architecture including layers, skip connections, and concatenations. At this stage, the contrast-enhanced input generated earlier is patched around the tumour region. This patch is then fed into the modified U-Net architecture, which produces the final segmentation output. Finally, the segmented patches are reassembled and mapped back to the size of the original image, providing a complete and precise tumour segmentation result.

minimized. The lightweight U-Net architecture then performs precise segmentation on these patches. Finally, the segmented patches are reassembled and mapped back to the size of the original image, providing a complete and precise tumour segmentation result. The integration of YOLOv3 and the attention-enhanced U-Net architecture enables faster and more efficient tumour localization and segmentation while maintaining computational efficiency with high accuracy. This hybrid framework provides an optimized solution for brain tumour detection and segmentation in brain MR imaging workflows. The architecture of the proposed network is shown in Figure 3.

A. Dataset and Data Pre-Processing

1) Dataset: This brief utilizes the Figshare brain MRI dataset, sourced from two medical institutions: Nanfang Hospital and General Hospital, Tianjin Medical University, China [6]. The dataset contains 3064 T1-weighted MRI slices from 233 patients diagnosed with one of three types of brain tumours: 708 slices of meningiomas, 1426 slices of gliomas, and 930 slices of pituitary tumours. These slices cover various imaging planes, including sagittal, coronal, and axial. Figure 1 illustrates different tumour types of gliomas, meningiomas and pituitary with different planes from the dataset. The MRI scans in the dataset have a resolution of 512×512 with each pixel corresponding to a size of $0.49 \times 0.49\ mm^2$. Tumour borders were manually annotated by three experienced radiologists. Within the dataset, each MRI scan is associated with a unique Patient Identification (Patient ID), ensuring proper and distinct identification for each patient's data. Each MRI slice in the dataset is associated with a tumour type label, denoted as l_{gt}, which indicates whether the tumour is a meningioma, glioma, or pituitary tumour. In addition, each slice has a corresponding tumour mask, denoted as T_{ij}. The pairing of $\{l_{gt}, T_{ij}\}$ provides the ground truth segmentation for the tumour region during the training process.

2) Data Pre-Processing: Pre-processing was performed on the input MRI images to reduce the dimensionality of the input image, thereby minimizing computational complexity and improving the efficiency of the segmentation process. Each original MR image, with dimensions of 512×512, was resized to 256×256 without distortion, effectively reducing the total number of pixels. Additionally, Contrast Limited Adaptive Histogram Equalization (CLAHE) [17] was applied to the MRI images as an image enhancement technique. CLAHE helps to mitigate noisy elements and enhances the visible features within the images, making them more suitable for training. To further bolster the robustness of the network and reduce the risk of overfitting, several data augmentation techniques were employed, including random flipping, random rotations, and 90-degree clockwise rotations. These transformations increase the diversity of the training data. The pre-processed and augmented MRIs are subsequently fed into the proposed architecture to enhance feature learning and optimize segmentation effectiveness.

B. Network Architecture

1) Proposed Architecture: This section outlines the implementation of the proposed architecture for brain tumour segmentation. The architecture takes a contrast-enhanced input, which is generated by computing the dot product between the output from YOLOv3 and the original image, then adding the result back to the original image which is described in Section II. The attention mechanism, which includes both channel attention and spatial attention, is one of the key approaches in image processing tasks. In this brief, a 2D U-Net architecture is employed, incorporating both channel and spatial attention mechanisms to enhance segmentation performance. The proposed attention module operates in parallel, with channel and spatial attention integrated alongside a skip connection. This integration of skip connections helps to reduce redundancy and sparsity in the architecture. A visual representation of the detailed architecture can be found in Figure 3.

2) Neural Network Training and Implementation: In this brief, the model was implemented using Python 3.10.12, with TensorFlow and the Keras library. The dataset has been divided into three parts in order to train the deep learning model, 70% for training, 15% for validation, and 15% for testing. The data was partitioned based on patient IDs to account for potential correlations between multiple MRI images of the same patient taken from different angles. The MRI data was collected from a total of 233 patients, and the corresponding MR images were retrieved using the Patient IDs provided in the dataset. 70% of the patients (163 patients), which corresponded to

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(2160 images) of the total MR images, were assigned to a training set. Additionally, 15% of the patients (35 patients), representing (452 images) of the total dataset, were used for validation, and the remaining 15% of the patients (35 patients), corresponding to (452 images), were reserved for performance evaluation. For implementation, each MRI image was resized to dimensions of 256 × 256. Several data augmentation techniques were applied during training to enhance the model's robustness. To optimize training efficiency, a patch corresponding to the bounding box coordinates obtained from Step 1, which outlines the tumorous area, was used as the input for the training process. The model was trained for 100 epochs with a batch size of 32. The loss function used in this brief was the weighted cross-entropy loss, inspired by Ho and Wookey [18]. The optimization was performed using the Stochastic Gradient Descent (SGD) algorithm, with an initial learning rate of 0.001 and a momentum coefficient of 0.9. To evaluate the performance of the deep learning model, 5-fold cross-validation was employed using the testing dataset. The evaluation metrics were analyzed to measure the segmentation effectiveness and reliability.

III. Experimental Results

A. Performance Evaluation

This section presents the results, analysis and performance evaluation of the proposed method. For performance evaluation, the segmentation results are assessed on the test dataset by calculating key metrics: precision, recall, Jaccard index, and Dice coefficient. The model achieved 0.9127, 0.8974, 0.9273, and 0.8915 for the following metrics respectively. The results show that the proposed method provides us with satisfactory results in image segmentation. One notable observation is the close proximity of the precision and recall values, which suggests that the proposed method is robust to the pixel imbalance between tumourous and non-tumourous regions. This indicates that the model effectively handles the inherent challenges posed by class imbalance in medical imaging datasets. The results indicate that using MRI patches instead of the entire image as input may improve computational efficiency and segmentation performance. While this approach is expected to be more computationally efficient, the precise quantification of computational cost reduction is beyond the scope of this brief. Figure 4 provides a visual comparison between the true label and the predictions generated by the proposed method: (a) Input image, (b) Ground truth, (c) enhanced input image and (d) Prediction of the Proposed Model. This figure highlights the model's ability to produce accurate segmentation results that align closely with the ground truth annotations.

B. Comparison With Literature

The performance of the proposed method was evaluated by comparing it with existing approaches from the literature, employing various DL methods and experimental setups. The comparison was conducted against state-of-the-art methods. As represented in Tables I the following Dice coefficients for each method have been obtained: ResU-Net [20] achieved a Dice coefficient of 0.8751. Sheela and Suganthi [21] reported a Dice of 0.631. Allah et al. [22], utilizing the Edge U-Net architecture, achieved an average Dice score of 0.892. Shafai et al. [23] reported an average Dice score of 0.3518 and a Jaccard index of 0.2134 but did not provide scores for individual tumour types. VGG19-U-Net [24] attained Dice scores of 0.81 and 0.86 for enhancing tumours and core tumours, respectively. The multi-pathway CNN architecture proposed by Díaz-Pernas et al. [25] achieved a Dice score of 0.828. In comparison, our proposed model obtained a Dice score of 0.8915, demonstrating competitive performance among these methods. As presented in Tables II and III, the Jaccard index and Recall were compared across the listed

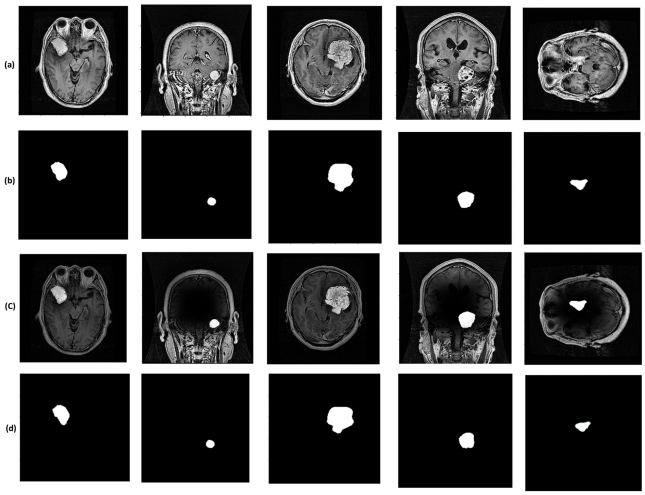

Fig. 4. Segmentation results with a subset of five MRI scans, (a) Input image, (b) Ground truth, (c) Enhanced Image, (d) Prediction of Proposed Method.

TABLE I
Dice Scores for Various DL Models for MRI Tumor Segmentation

Methodologies	Dice Score			
	Glioma	Menejioma	Pituitary	Avg.
U-Net [19]	0.8519	0.8592	0.8520	
Res U-Net [20]	0.8627	0.8916	0.8715	0.8751
Sheela et al. [21]	0.59	0.78	0.49	0.631
Edge U-Net network [22]	0.9176	0.888	0.8728	0.892
El Shafai et al. [23]	–	–	–	0.3518
VGG U-Net [24]	0.8033	–	0.86	0.831
Multi-Pathway CNN [25]	0.779	0.894	0.813	0.828
Proposed Method	**0.9214**	**0.8994**	**0.8805**	0.8915

TABLE II
Jaccard Index for Various DL Models Using the Figshare Dataset

Methodologies	Jaccard Index			
	Glioma	Menejioma	Pituitary	Avg.
ResU-Net [20]	0.7419	0.7532	0.7422	0.7457
El Shafai et al. [23]	–	–	–	0.2134
Edge U-Net network [22]	0.8478	0.7985	0.7743	0.805
Proposed Method	**0.8754**	**0.8953**	**0.8792**	**0.9273**

TABLE III
Recall Performance Across DL Models for MRI Tumor

Methodologies	Recall			
	Glioma	Menejioma	Pituitary	Avg.
Sheela et al. [21]	0.51	0.67	0.44	
El Shafai et al. [23]	–	–	–	0.3792
Multi-Pathway CNN [25]	0907	**0.961**	**0.954**	**0.940**
Proposed Method	**0.9214**	0.8994	0.8805	0.8974

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TABLE IV
PERFORMANCE EVALUATION ACROSS TUMOR TYPES: GLIOMA, MENINGIOMA, AND PITUITARY TUMORS

Tumour Type	Dice	Jaccard	Precision	Recall
Glioma	0.9102	0.8754	0.8916	0.9214
Menejioma	0.8976	0.8953	0.8972	0.8994
Pituitary	0.8914	0.8792	0.8868	0.8805

approaches. Additionally, Table IV highlights the performance evaluation of the proposed method for different tumour types: Glioma, Meningioma, and Pituitary tumours. Furthermore, in some other studies within the literature, such as the one presented by Chouksey et al. [26], where their evaluation was conducted on the Figshare dataset [6] and extended to other databases. However, their reported results include metrics and segmentation outcomes for only a small subset of seven images out of 3064 MRI scans. Similarly, Kaldera et al. [27] introduced a neural network for brain tumour segmentation and classification for various tumour types with limited MRI scans. However, their segmentation results are provided with only a limited number of examples, making them unsuitable for comparison. It has been shown that the proposed architecture, among other methods, obtains a higher average dice index value of 0.8915 and achieves better results than the results presented in the literature. While the proposed method demonstrates promising segmentation performance, a direct comparison of computational complexity with other approaches is not included due to differences in implementation settings, parameter configurations, and hardware resources. Acknowledging the importance of computational efficiency, a standardized evaluation of computational cost across comparable methods remains an important direction for future work.

IV. CONCLUSION

This brief introduces a novel approach for detecting and segmenting brain tumours in MRI (CE-MRI) images. The proposed framework begins by employing YOLOv3 for tumour localization, identifying the ROIs within the MRI scans. After localization, the method leverages a modified U-Net architecture to precisely segment the tumour boundaries. This two-stage approach allows for efficient and accurate segmentation by first detecting the tumour location and then refining the segmentation with a modified U-NET architecture. The evaluation was conducted using the Figshare brain dataset. The results show promising performance, with precision of 0.9127, recall of 0.8974, a Jaccard index of 0.9273, and a Dice coefficient of 0.8915. The proposed method outperforms existing techniques in the literature, demonstrating its potential for efficient and accurate brain tumour segmentation.

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